PADHAKU
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CHAPTER 3

The World of Numbers

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples. write your own.

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📝 Exercise Set 3.1

1 **A merchant in Lothal receives 15 copper ingots for every 2 bags of spices. If he brings 12 bags, how many ingots will he leave with?**

✔ ANSWER

12 bags = 12 ÷ 2 = 6 groups of 2 bags

Each group gives 15 ingots

Ingots = 6 × 15 = **90 copper ingots**

2 **Ishango bone column: 11, 13, 17, 19. What do these numbers have in common? List the next three numbers that fit this pattern.**

✔ ANSWER

All of them are **prime numbers** (only factors are 1 and the number itself). They are the primes between 10 and 20.

Next three primes: **23, 29, 31**

3 **Natural numbers are closed under addition. Are they closed under subtraction? Give examples.**

✔ ANSWER

No, natural numbers are not closed under subtraction.

- 3 − 5 = −2, which is not a natural number.
- 4 − 4 = 0, which is not a natural number.

(Sometimes it works, e.g. 7 − 2 = 5, but not always.)

***4** **Each finger has 3 joints and the thumb counts them. How many can you count on one hand? How does this relate to base-12 counting?**

💡 **SAMPLE ANSWER**

4 fingers (not the thumb) × 3 joints = **12** on one hand.

So one hand naturally counts a **dozen (12)**. This is why base-12 systems (dozens, 12 months, 12 hours) were easy to use.

Using the 5 fingers of the other hand to count how many dozens: 5 × 12 = 60 (base-60 counting of minutes and seconds).

🤔 Think and Reflect · 3.3.1 The Arithmetic of Integers

? **Why does a negative times a negative equal a positive? Think of it in terms of action and debt.**

✔ **ANSWER**

A negative number is a debt; multiplying by a negative number means **removing** debts.

If someone takes away 4 of your debts of ₹3 each, you are ₹12 richer.

So (−3) × (−4) = **+12**.

📝 Exercise Set 3.2

1 **Temperature in Ladakh is 4 °C at noon. By midnight it drops by 15 °C. What is the midnight temperature?**

✔ **ANSWER**

Midnight temperature = 4 − 15

= **−11 °C**

2 **A spice trader takes a loan of ₹850, then makes a profit of ₹1,200, then a loss of ₹450. Write this as an equation using integers and find his final standing.**

✔ ANSWER

Debt = −850, profit = +1200, loss = −450

(−850) + 1200 + (−450)

= 350 − 450 = −100

Final standing: **−₹100**, i.e. he is in a debt of ₹100.

3 **Calculate using Brahmagupta's laws: (i) (−12) × 5 (ii) (−8) × (−7) (iii) 0 − (−14) (iv) (−20) ÷ 4**

✔ ANSWER

(i) debt × fortune = debt: (−12) × 5 = **−60**

(ii) debt × debt = fortune: (−8) × (−7) = **56**

(iii) 0 − (−14) = 0 + 14 = **14**

(iv) (−20) ÷ 4 = **−5**

4 **Explain, using a real-world example of debt, why subtracting a negative number is the same as adding a positive number (e.g., 10 − (−5) = 15).**

💡 SAMPLE ANSWER

Suppose you have ₹10 but also owe a friend ₹5 (a debt of −5).

If your friend says "forget the ₹5", the debt is taken away (subtracted).

Removing a debt of ₹5 makes you ₹5 better off: you now have ₹10 and owe nothing, worth ₹15 compared to before.

So 10 − (−5) = 10 + 5 = **15**. Taking away a debt is the same as adding money.

🤔 Think and Reflect · 3.4 Fractions and Rational Numbers

? **Can you explain why we need q ≠ 0 in the definition of a rational number?**

✔ ANSWER

p/q means p ÷ q. Division by 0 has no meaning: no number times 0 gives a non-zero p (and for p = 0 every number works).

So p/0 is not defined, and we need **q ≠ 0**.

1 **While adding or subtracting two rational numbers having different denominators, how will you make the denominators equal?**

✔ ANSWER

Find the **LCM** of the two denominators.

Change each fraction to an equivalent fraction with the LCM as denominator (multiply numerator and denominator by the same number).

Example: 1/4 + 1/6 → LCM 12 → 3/12 + 2/12 = 5/12.

2 **Verify the distributive law for rational numbers.**

💡 SAMPLE ANSWER

Take p = 1/2, q = 1/3, r = 1/4.

LHS: p(q + r) = 1/2 × (4/12 + 3/12) = 1/2 × 7/12 = 7/24

RHS: pq + pr = 1/6 + 1/8 = 4/24 + 3/24 = 7/24

LHS = RHS, so **p(q + r) = pq + pr** holds.

📝 Exercise Set 3.3

1 **Prove that the following rational numbers are equal: (i) 2/3 and 4/6 (ii) 5/4 and 10/8 (iii) −3/5 and −6/10**

(iv) 9/3 and 3

✔ ANSWER

a/b = c/d when ad = bc.

(i) 2 × 6 = 12 and 3 × 4 = 12 → **equal**

(ii) 5 × 8 = 40 and 4 × 10 = 40 → **equal**

(iii) (−3) × 10 = −30 and 5 × (−6) = −30 → **equal**

(iv) 3 = 3/1; 9 × 1 = 9 and 3 × 3 = 9 → **equal**

2 **Find the sum: (i) 2/5 + 3/10 (ii) 7/12 + 5/8 (iii) −4/7 + 3/14**

✔ ANSWER

(i) 4/10 + 3/10 = **7/10**

(ii) LCM 24: 14/24 + 15/24 = **29/24**

(iii) −8/14 + 3/14 = **−5/14**

3 **Find the difference: (i) 5/6 − 1/4 (ii) 11/8 − 3/4 (iii) −7/9 − (−2/3)**

✔ ANSWER

(i) 10/12 − 3/12 = **7/12**

(ii) 11/8 − 6/8 = **5/8**

(iii) −7/9 + 2/3 = −7/9 + 6/9 = **−1/9**

4 **Find the product: (i) 2/3 × 3/10 (ii) 7/11 × 5/8 (iii) −4/7 × 5/14**

✔ ANSWER

(i) 6/30 = **1/5**

(ii) **35/88**

(iii) −20/98 = **−10/49**

5 **Find the quotient: (i) 2/3 ÷ 3/10 (ii) 7/11 ÷ 5/8 (iii) −4/7 ÷ 5/14**

✔ ANSWER

(i) 2/3 × 10/3 = **20/9**

(ii) 7/11 × 8/5 = **56/55**

(iii) −4/7 × 14/5 = −56/35 = **−8/5**

6 **Show that: (1/2 + 3/4) × 8/3 = 1/2 × 8/3 + 3/4 × 8/3**

✔ ANSWER

LHS: 1/2 + 3/4 = 2/4 + 3/4 = 5/4

5/4 × 8/3 = 40/12 = 10/3

RHS: 1/2 × 8/3 = 4/3; 3/4 × 8/3 = 2

4/3 + 2 = 4/3 + 6/3 = 10/3

LHS = RHS = **10/3**. Hence shown.

7 **Simplify using the distributive property: 7/9 × (6/7 − 3/4)**

✔ ANSWER

= 7/9 × 6/7 − 7/9 × 3/4

= 6/9 − 21/36

= 2/3 − 7/12 = 8/12 − 7/12

= **1/12**

8 **Find the rational number x such that: 5/6 × (x + 3/5) = 5/6 x + 1/2**

✔ ANSWER

LHS = 5/6 x + 5/6 × 3/5 (distributive law)

5/6 × 3/5 = 15/30 = 1/2

So LHS = 5/6 x + 1/2, which is exactly the RHS.

The equation is true for every x: **any rational number x** satisfies it (e.g. x = 0, 1, −2/3...).

🤔 Think and Reflect · 3.4.1 Representation of Rational Numbers on the Number Line

? Try and represent 8/5 and −7/4 on a number line.

✔ ANSWER

8/5 = 1 3/5: it lies between 1 and 2. Divide the gap from 1 to 2 into 5 equal parts and move 3 parts to the right of 1.

−7/4 = −1 3/4: it lies between −1 and −2. Divide the gap from −1 to −2 into 4 equal parts and move 3 parts to the left of −1.

⭐ 3.4.2 The Density of Rational Numbers

? Explain why the average of two rational numbers a and b, (a + b)/2, is always a rational number between a and b.

✔ ANSWER

Rational: a + b is rational (closure under addition), and dividing by 2 (non-zero) keeps it rational.

Between: let a < b.

(a + b)/2 − a = (b − a)/2 > 0, so (a + b)/2 > a

b − (a + b)/2 = (b − a)/2 > 0, so (a + b)/2 < b

So **a < (a + b)/2 < b**; it is the midpoint of a and b.

? Rational numbers are dense. Do they completely fill the number line?

✔ ANSWER

No. There are gaps: numbers like √2 and π are on the number line but are not rational (see 3.5).

📝 Exercise Set 3.4

1 **Represent the rational numbers 2/3, −5/4 and 1½ on a single number line.**

✔ ANSWER

Draw a number line with −2, −1, 0, 1, 2.

- **2/3**: divide 0 to 1 into 3 equal parts; mark the 2nd point to the right of 0.
- **−5/4** = −1 1/4: divide −1 to −2 into 4 equal parts; mark the 1st point to the left of −1.
- **1½** = 3/2: the midpoint of 1 and 2.

2 **Find three distinct rational numbers that lie strictly between −1/2 and 1/4.**

✔ ANSWER

−1/2 = −4/8 and 1/4 = 2/8

Numbers between: −3/8, −2/8, −1/8, 0, 1/8

Three of them: **−1/4, 0, 1/8**

3 **Simplify: (−1/4) + (5/12)**

✔ ANSWER

= −3/12 + 5/12

= 2/12 = **1/6**

4 **A tailor has 15¾ metres of silk. One kurta needs 2¼ metres. Exactly how many kurtas can he make?**

✔ ANSWER

$15\frac{3}{4} = 63/4$, $2\frac{1}{4} = 9/4$

$63/4 \div 9/4 = 63/4 \times 4/9 = 63/9 = 7$

He can make exactly **7 kurtas** (no silk left).

5 **Find three rational numbers between 3.1415 and 3.1416.**

✔ ANSWER

Write them as 3.14150 and 3.14160.

Numbers between: **3.14151, 3.14153, 3.14155** (any of 3.14151 ... 3.14159).

***6** **Can you think of other way(s) to find a rational number between any two rational numbers?**

💡 SAMPLE ANSWER

- **Common denominator method:** write both with the same denominator, multiply numerator and denominator by 10 (or more), then pick the fractions in between. E.g. 1/3, 1/2 → 20/60, 30/60 → 21/60, 22/60 ...
- **Decimal method:** write both as decimals and choose a decimal in between.
- **Mediant:** for $a/b < c/d$ ($b, d > 0$), $(a + c)/(b + d)$ lies between them. E.g. between 1/3 and 1/2: 2/5.
- **$a + (b - a)/n$** for any $n > 1$ also lies between a and b.

🤔 Think and Reflect · 3.5 Irrational Numbers

? **Can $\sqrt{2}$ be written as a rational number p/q?**

✔ ANSWER

No. √2 cannot be written as p/q with integers p, q (q ≠ 0). It is irrational (proof in 3.5.1).

? **Try to prove the irrationality of √3 using proof by contradiction. Will the same approach work for √5, √7 or √10?**

✔ ANSWER

1. Assume √3 = p/q in simplest form (p, q co-prime, q ≠ 0).

2. Squaring: $3 = p^2/q^2$, so $3q^2 = p^2$.

3. 3 divides p^2, and 3 is prime, so 3 divides p. Let p = 3k.

4. $3q^2 = 9k^2$, so $q^2 = 3k^2$. Then 3 divides q^2, so 3 divides q.

5. 3 divides both p and q: contradiction with simplest form.

6. So **√3 is irrational**.

Yes, the same method works for √5 and √7 (5 and 7 are primes). For √10: $p^2 = 10q^2$ is even, so p = 2k; then $4k^2 = 10q^2$, $2k^2 = 5q^2$, so q is even too: contradiction. So √10 is also irrational. (It works for √n whenever n is not a perfect square.)

? **We have seen how to obtain a line whose length is a rational number. How do we obtain lines whose lengths are irrational?**

✔ ANSWER

Use right triangles and the **Baudhāyana–Pythagoras theorem**. A right triangle with legs 1 and 1 has hypotenuse $\sqrt{(1^2 + 1^2)} = \sqrt{2}$. Then a compass carries this length to the number line.

? **Extend this method to construct line segments of lengths √3 and √5 using a ruler and a compass. Generalise to √n.**

✔ ANSWER

√3: from Fig. 3.11 we have OB = √2. At B draw BC ⊥ OB with BC = 1. Then OC = √(2 + 1) = √3. With centre O and radius OC draw an arc to cut the number line: that point is √3.

√5: mark OA = 2 on the number line, draw AB ⊥ OA with AB = 1. Then OB = √(4 + 1) = √5; carry it to the line with a compass.

General √n: having a segment of length √(n − 1), draw a perpendicular of length 1 at its end; the hypotenuse is √(n − 1 + 1) = √n. Repeating this step gives √2, √3, √4, √5 … (the square root spiral).

⭐ 3.6 Real Numbers: Decimals and Cyclic Patterns

? **Example 2: 3/8 = 0.375. Can you tell for which rational numbers the decimal will be terminating?**

✔ ANSWER

Those whose denominator (in lowest form) has **no prime factors other than 2 and 5**, e.g. $8 = 2^3$, $20 = 2^2 \times 5$.

? **Try to find the decimal expansions of 10/3 and 11/12. What do you observe about the repetition of the digits after the decimal point?**

✔ ANSWER

10/3 = 3.333… = $\mathbf{3.\overline{3}}$

11/12 = 0.91666… = $\mathbf{0.91\overline{6}}$

In 10/3 the digit 3 repeats right after the decimal point. In 11/12 the digits 9 and 1 do not repeat, and then 6 repeats forever.

? **Dividing 1 by 7, the possible remainders are 1, 2, 3, 4, 5, 6. Why not 0?**

✔ ANSWER

A remainder 0 would end the division, so 1/7 would be a terminating decimal $1/7 = m/10^k$, i.e. 7 would divide 10^k.

But 10^k has only prime factors 2 and 5, so 7 never divides it. So the remainder is **never 0**.

? **The decimal expansion of p/q will be terminating precisely when the prime factors of q are only 2, only 5, or both 2 and 5. Can you explain why?**

✔ ANSWER

If $q = 2^m \times 5^n$, multiply numerator and denominator by enough 2s or 5s to make the denominator a **power of 10**. E.g. $3/20 = 3/(2^2 \times 5) = 15/100 = 0.15$. A fraction over 10^k is a terminating decimal.

If q (in lowest form) has another prime factor (like 3 or 7), it can never divide a power of 10, so no remainder becomes 0 and the decimal repeats.

📝 Exercise Set 3.5

1 **Without long division, determine which of 7/20, 4/15 and 13/250 have terminating decimals and which are repeating. Then check by long division.**

✔ ANSWER

- 7/20: $20 = 2^2 \times 5$ → **terminating**. 7/20 = 35/100 = **0.35**
- 4/15: $15 = 3 \times 5$ (has 3) → **repeating**. 4/15 = 0.2666... = $\mathbf{0.2\overline{6}}$
- 13/250: $250 = 2 \times 5^3$ → **terminating**. 13/250 = 52/1000 = **0.052**

2 **Perform the long division for 1/13. Identify the repeating block. Does it show cyclic properties if you evaluate 2/13? Now compute 3/13, 4/13, etc. What do you notice?**

✔ ANSWER

1/13 = 0.076923076923... = 0.076923 repeating; block **076923** (6 digits).

2/13 = 0.153846... block **153846**: this is **not** a rotation of 076923, so 2/13 does not show the cyclic property.

Fraction	Repeating block	Family
1/13	076923	A
2/13	153846	B
3/13	230769	A
4/13	307692	A
5/13	384615	B
6/13	461538	B
7/13	538461	B
8/13	615384	B
9/13	692307	A
10/13	769230	A
11/13	846153	B
12/13	923076	A

Observation: the blocks form two cycles. 1, 3, 4, 9, 10, 12 (×1/13) give rotations of 076923; 2, 5, 6, 7, 8, 11 give rotations of 153846. Unlike 1/7, the block of 1/13 is **not fully cyclic** (it has only 6 digits for 12 multiples).

3 **Classify as rational or irrational, and find the fractions of the rational ones: (i) √81 (ii) √12 (iii) 0.33333... (iv) 0.123451234512345... (v) 1.01001000100001... (vi) 23.5601856122398747901 20**

✔ ANSWER

(i) √81 = 9 = **9/1** → rational

(ii) √12 = 2√3 → **irrational** (12 is not a perfect square)

(iii) x = 0.333..., 10x = 3.333..., 9x = 3 → x = **1/3** → rational

(iv) block 12345 repeats (5 digits): 100000x − x = 12345 → x = 12345/99999 = **4115/33333** → rational

(v) The number of zeros keeps increasing (1, 2, 3, 4 ...); no single block repeats and it does not end → **irrational**

(vi) terminating decimal (21 decimal places) → rational: $23560185612239874790120/10^{21}$ = **589004640305996869753/25000000000000000000**

4 **Using algebra (let x = $0.\overline{9}$, multiply by 10 and subtract), explain why $0.\overline{9} = 1$.**

✔ ANSWER

Let x = 0.999...

10x = 9.999...

10x − x = 9.999... − 0.999...

9x = 9

x = 1, so $0.\overline{9} = 1$ exactly.

***5** **The repeating block of 1/7 is a cyclic number. Try to find more numbers n whose reciprocals 1/n produce decimals with repeating blocks that are cyclic.**

💡 **SAMPLE ANSWER**

We need 1/n with a block of length n − 1 (the longest possible). Examples: **n = 17, 19, 23, 29**.

1/17 = 0.0588235294117647 repeating (16 digits). Multiplying this block by 1 to 16 only rotates its digits.

1/19 has an 18-digit cyclic block: 052631578947368421.

(Such n are primes; other examples: 47, 59, 61, 97.)

🤔 Think and Reflect · 3.7 Conclusion

? **What is the square root of −1? Is there a real number that, when multiplied by itself, gives a negative number?**

✔ **ANSWER**

No real number works: a positive × positive is positive and a negative × negative is also positive (and 0 × 0 = 0).

So √−1 is not on the real number line. Mathematicians defined a new number **i** (imaginary unit) with i × i = −1.

📝 End-of-Chapter Exercises

1 **Convert to a terminating or non-terminating repeating decimal by long division: (i) 3/50 (ii) 2/9**

✔ **ANSWER**

(i) 3 ÷ 50 = **0.06** (terminating)

(ii) 2 ÷ 9 = 0.222... = **$0.\overline{2}$** (non-terminating repeating; remainder 2 keeps coming back)

2 **Prove that √5 is an irrational number.**

✔ ANSWER

1. Assume √5 is rational: √5 = p/q, with p, q co-prime integers, q ≠ 0.

2. Squaring: 5 = p^2/q^2, so $5q^2 = p^2$.

3. So 5 divides p^2. As 5 is prime, 5 divides p. Let p = 5k.

4. Then $5q^2 = 25k^2$, so $q^2 = 5k^2$.

5. So 5 divides q^2, hence 5 divides q.

6. Now 5 is a common factor of p and q: contradiction (they were co-prime).

7. So our assumption is wrong: **√5 is irrational**.

3 **Convert into the form p/q: (i) 12.6 (ii) 0.0120 (iii) $3.05\overline{2}$ (iv) $1.2\overline{35}$ (v) $0.\overline{23}$ (vi) $2.0\overline{5}$ (vii) $2.12\overline{5}$ (viii) $3.1\overline{25}$ (ix) $2.\overline{1625}$**

✔ ANSWER

(i) 12.6 = 126/10 = **63/5**

(ii) 0.0120 = 120/10000 = **3/250**

(iii) x = 3.05252...; 10x = 30.5252...; 1000x = 3052.5252...

990x = 3022 → x = 3022/990 = **1511/495**

(iv) x = 1.23535...; 10x = 12.3535...; 1000x = 1235.3535...

990x = 1223 → x = **1223/990**

(v) x = 0.2323...; 100x = 23.2323...; 99x = 23 → x = **23/99**

(vi) x = 2.0555...; 10x = 20.555...; 100x = 205.555...

90x = 185 → x = 185/90 = **37/18**

(vii) x = 2.12555...; 100x = 212.555...; 1000x = 2125.555...

900x = 1913 → x = **1913/900**

(viii) x = 3.12555...; 100x = 312.555...; 1000x = 3125.555...

900x = 2813 → x = **2813/900**

(ix) x = 2.16251625...; 10000x = 21625.1625...

9999x = 21623 → x = **21623/9999**

4 **Locate on the number line: (i) 0.532 (ii) $1.15\overline{}$**

✔ ANSWER

(i) 0.532 lies between 0.5 and 0.6. Divide 0.5 to 0.6 into 10 parts: it lies between 0.53 and 0.54. Divide 0.53 to 0.54 into 10 parts: **0.532 is the 2nd mark after 0.53**.

(ii) $1.15\overline{}$ = 1.1555... = 52/45 = 1 + 7/45.

It lies between 1.15 and 1.16; dividing that gap into 10 parts, it lies between 1.155 and 1.156, very close to 1.156.

Exact way: divide 1 to 2 into 45 equal parts and **mark the 7th point after 1**.

5 **Find 6 rational numbers between 3 and 4.**

✔ ANSWER

3 = 30/10, 4 = 40/10

31/10, 32/10, 33/10, 34/10, 35/10, 36/10 (i.e. 3.1, 3.2 ... 3.6)

6 **Find 5 rational numbers between 2/5 and 3/5.**

✔ ANSWER

Multiply numerator and denominator by 6: 2/5 = 12/30, 3/5 = 18/30

13/30, 14/30, 15/30, 16/30, 17/30

7 **Find 5 rational numbers between 1/6 and 2/5.**

✔ ANSWER

LCM of 6 and 5 = 30: 1/6 = 5/30, 2/5 = 12/30

6/30, 7/30, 8/30, 9/30, 10/30 (i.e. 1/5, 7/30, 4/15, 3/10, 1/3)

8 **If x/3 + x/5 = 16/15, find x.**

✔ ANSWER

x/3 + x/5 = 5x/15 + 3x/15 = 8x/15

8x/15 = 16/15

8x = 16 → **x = 2**

9 **a and b are non-zero rational numbers with a + 1/b = 0. Without numerical values, is ab positive or negative?**

✔ ANSWER

a + 1/b = 0 → a = −1/b

Multiply both sides by b: ab = −1

So **ab is negative** (ab = −1).

(Also: a and −1/b are equal, so a and b have opposite signs.)

10 **A rational number has a terminating decimal whose last non-zero digit is in the 4th decimal place. Show it can be written as $p/10^4$, p an integer not divisible**

by 10. Must its denominator in lowest form be divisible by 2^4 or 5^4?

✔ ANSWER

Let the number be $x = n.d_1d_2d_3d_4$ with $d_4 \neq 0$.

Then $10^4 \times x = p$ is an integer whose last digit is $d_4 \neq 0$, so **p is not divisible by 10**, and $x = p/10^4$.

$10^4 = 2^4 \times 5^4$. Since $10 \nmid p$, p is not divisible by both 2 and 5: either $2 \nmid p$ or $5 \nmid p$.

- If $2 \nmid p$, no factor 2 cancels, so the lowest-form denominator still has 2^4.
- If $5 \nmid p$, no factor 5 cancels, so it still has 5^4.

Yes, the denominator in lowest form is divisible by 2^4 or 5^4. Example: 0.0125 = 125/10000 = 1/80, and $80 = 2^4 \times 5$.

11 **Without division, is 18/125 terminating? If yes, how many decimal places?**

✔ ANSWER

$125 = 5^3$ (only the prime 5) → **terminating**.

18/125 = (18 × 8)/(125 × 8) = 144/1000 = 0.144

It has **3 decimal places**.

12 **A rational number in lowest form has denominator $2^3 \times 5$. How many decimal places will its decimal expansion have? Explain.**

✔ ANSWER

Denominator $2^3 \times 5 = 40$. Multiply numerator and denominator by $5^2 = 25$: denominator becomes $2^3 \times 5^3 = 10^3 = 1000$.

So the number = (25p)/1000. As p is co-prime to 40, p is odd and not a multiple of 5, so 25p ends in 5 (not 0).

So the decimal ends exactly at the 3rd place: **3 decimal places**. (In general, the number of places = the larger power of 2 or 5.)

***13** **$a = 7/12$, $b = 5/6$. Write $a = k_1/m$, $b = k_2/m$ with integers and $k_2 - k_1 > 6$. Write five rational numbers between a and b with denominator m. Explain why $k_2 - k_1 > n + 1$ is needed for n such numbers.**

✔ ANSWER

With m = 12: 7/12 and 10/12, difference 3 (too small). Take **m = 36**: a = 21/36, b = 30/36; $k_2 - k_1 = 9 > 6$.

Five numbers: **22/36, 23/36, 24/36, 25/36, 26/36**

Reason: with the same denominator m, the numbers between are k/m for the integers k strictly between k_1 and k_2. There are only $k_2 - k_1 - 1$ such integers.

To get n numbers we need $k_2 - k_1 - 1 \geq n$, i.e. $k_2 - k_1 \geq n + 1$. The condition $k_2 - k_1 > n + 1$ makes sure there are enough integers (more than n) between k_1 and k_2. If the gap is too small, multiply numerator and denominator by a bigger number to widen it.

***14** **$x + y + z = 0$ and $xy + yz + zx = 0$ for rational x, y, z. Show that x, y, z must all be zero.**

✔ ANSWER

$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$

$0^2 = x^2 + y^2 + z^2 + 2 \times 0$

So $x^2 + y^2 + z^2 = 0$.

Each square is ≥ 0, and their sum is 0, so each square is 0.

Hence **x = y = z = 0**.

***15** **Show that the rational number (a + b)/2 lies between the rational numbers a and b.**

✔ ANSWER

Let $a < b$ (so $b - a > 0$).

$(a + b)/2 - a = (b - a)/2 > 0 \rightarrow (a + b)/2 > a$

$b - (a + b)/2 = (b - a)/2 > 0 \rightarrow (a + b)/2 < b$

So **$a < (a + b)/2 < b$**. It is rational because sums and quotients (by 2) of rationals are rational.

16 **Find the lengths of the hypotenuses of all the right triangles in Fig. 3.14 (square root spiral).**

✔ ANSWER

Each new triangle has one leg = previous hypotenuse and other leg = 1.

1st: $\sqrt{(1^2 + 1^2)} = \sqrt{2}$

2nd: $\sqrt{((\sqrt{2})^2 + 1^2)} = \sqrt{3}$, then $\sqrt{4} = 2$, $\sqrt{5}$, ...

The figure has 10 right triangles, so the hypotenuses are:

$\sqrt{2}$, $\sqrt{3}$, 2 (= $\sqrt{4}$), $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$, $2\sqrt{2}$ (= $\sqrt{8}$), 3 (= $\sqrt{9}$), $\sqrt{10}$, $\sqrt{11}$

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**